

Question 1

Which of the following is least likely to be included in the capital budgeting process?

- A. Incremental cash flows.
- B. Sunk costs.**
- C. Opportunity costs.

B is correct. Sunk costs cannot be avoided, and hence are not affected by the accept/reject decision. They are not included in the capital budgeting process and analysis. Uses of Capital

Question 2

Mubs invested in a fund which generated the following returns over the last four years:

Year	Assets under management at start of year (\$)	Net return (%)
1	45 million	25
2	60 million	20
3	50 million	-10
4	40 million	-8

The money weighted annual return is closest to:

- A. 6.75%.
- B. 5.58%.
- C. 9.00%.**

C is correct. All amounts are in million dollars. The table below shows the computation for cash flows at the start of every year:

Year	1	2	3	4
Balance from previous year	0	56.25	72	45
New investment at the start of the year (inflow)	45	$60 - 56.25 = 3.75$	0	0
Withdrawal at the start of the year (outflow)	0	0	$72 - 50 = 22$	$45 - 40 = 5$
New balance at the start of the year (given in the question)	45	60	50	40
Investment return for the year (given in the question)	25%	20%	-10%	-8%
Investment gain (loss)	$25\% * 45 = 11.25$	$20\% * 60 = 12$	$-10\% * 50 = -5$	$-8\% * 40 = -3.2$
Balance at the end of the year	$45 + 11.25 = 56.25$	$60 + 12 = 72$	$50 - 5 = 45$	$40 - 3.2 = 36.8$

In order to calculate the money-weighted return (IRR) we assume the final amount (36.8 million) is withdrawn. The money-weighted return is the IRR, which can be calculated using a financial calculator as follows:

CF0 = -45, CF1 = -3.75, CF2 = 22, CF3 = 5, CF4 = 36.8, CPT IRR.

IRR = 9.00%. Portfolio Risk and Return Part I; Section 2.1.

Question 3

A financial analyst is estimating the cost of capital of Acumen Corp. The following information is provided on primary competitors and their capital structures (in millions):

Competitor	Market Value of Debt	Market Value of Equity
A	63	101
B	87	113
C	51	119

If the analyst is estimating proportions of debt and equity using the company's competitors' capital structure, these proportions will be *closest* to:

- A. **wd=0.37, we=0.63**
- B. wd=0.38, we=0.62
- C. wd=0.35, we=0.65

A is correct. The proportions of debt and equity using competitors' capital structure can be calculated as,

$$w_d = \frac{\frac{63}{63+101} + \frac{87}{87+113} + \frac{51}{51+119}}{3} = 0.37$$
$$w_e = \frac{\frac{101}{63+101} + \frac{113}{87+113} + \frac{119}{51+119}}{3} = 0.63$$

Question 4

The benefits of algorithmic trading are:

- A. speed and anonymity but higher transaction costs.
- B. **best limit or market order and most appropriate trading venue.**
- C. seeking to earn a profit from investment decisions based on the end of day market prices.

B is correct. Algorithmic trading provides benefits of speed of execution, anonymity, and lower transaction costs. Algorithms may also determine the best way to price the order (e.g., limit or market order) and the most appropriate trading venue (e.g., exchange or dark pool) to route for execution. High-frequency trading (HFT), is a form of algorithmic trading which decides what to buy or sell and where to execute on the basis of real-time prices and market conditions, seeking to earn a profit from intraday market mispricings. Fintech; Section 6.4. LO.c.

Question 5

The operating breakeven point is the same as the breakeven point when the degree of financial leverage (DFL) is:

- A. -1.00.
- B. 0.00.
- C. **1.00.**

C is correct. When DFL = 1.00, operating income = net income. This implies zero debt. The breakeven point is: (fixed costs + fixed cost of debt) ÷ contribution margin. Because the fixed

cost of debt is zero, the company's breakeven point becomes: $\text{fixed costs} \div \text{contribution margin}$, which is the same as the operating breakeven point. Measures of Leverage; Section 3.4.

Question 6

A major supplier of Logs Limited has decided to increase its trade credit terms by fifteen days. This change is most likely to have the following effect on Logs Limited:

- A. Decrease in the company's cash conversion cycle.**
- B. Increase in the company's cash conversion cycle.
- C. Decrease in the company's operating cycle.

A is correct. An increase in trade credit terms by fifteen days will increase the number of days of payables. This will decrease the company's cash conversion cycle/net operating cycle. Note: the operating cycle does not consider days of payables; hence Option C is not correct. Sources of Capital

Question 7

Ahmer Khan manages a defined benefit plan which has a large number of retirees. The fund is most likely to have a high need for:

- A. liquidity.
- B. insurance.
- C. income.**

C is correct. Income is required to meet the cash flow obligations of retirees. Pension funds have a lower need for liquidity and do not need insurance. Portfolio Management Overview; Section 3.

Question 8

Which of the following issues discussed at a shareholders' general meeting would *most likely* require only a simple majority vote for approval?

- A. Approval of financial statements**
- B. Voting on a merger
- C. Amendments to bylaws

A is correct. Some ordinary resolutions, such as the approval of financial statements and the election of directors and auditors, require only a simple majority of votes to be passed. Special resolutions including amendments to bylaws, voting on a merger or takeover transaction, or waiving pre-emptive rights are more material in nature and, therefore, require a supermajority vote, such as two-thirds or 75% of votes, to be passed. *Introduction to Corporate Governance and Other ESG Considerations.*

Question 9

Risk budget is least likely measured through:

- A. evaluation of risks based on asset classes.
- B. risk measures such as standard deviation, beta and value at risk.
- C. setting a restriction on the amount of money to be spent on hedging strategies.**

C is correct. Risk budgeting quantifies tolerable risks by specific metrics and should drive hedging strategies. It can be measured in several ways. It may be a complex measure that evaluates risks based on their asset classes such as equity, commodities, real estate and then

allocates investment by their asset class. It may also be a simple risk measure such as standard deviation, beta, and value at risk and scenario loss.

Introduction to Risk Management; Section 3.3.

Question 10

FarmFresh Dairy has estimated the NPV of the expected cash flows from a new processing plant to be –USD 0.70 million. The company is evaluating an incremental investment of USD 0.35 million that would allow flexibility to switch to cheaper energy sources. The option to switch to cheaper sources of energy when they are available has an estimated value of USD 1.40 million. The value of the new processing plant including this real option to use alternative energy sources would be:

- A. USD 1.05 million
- B. USD 0.35 million**
- C. USD -2.45 million

B is correct.

The NPV, including the real option, should be:

Project NPV = NPV (based on DCF alone) – Cost of options + Value of options.

Project NPV = –0.70 million – 0.35 million + 1.40 million = USD 0.35 million. *Uses of Capital.*

Question 11

The following information about a private company and its comparable public entity has been gathered:

	Tax rate	Debt-to-equity ratio	Equity beta
Private company	40 percent	0.9	n/a
Public company	36 percent	0.5	1.76

The estimated equity beta for the private company, using the pure play method, is closest to:

- A. 1.33.
- B. 2.05.**
- C. 2.53

B is correct.

The asset beta for the public company:

$$1.76 / [1 + (1 - 0.36) (0.5)] = 1.33$$

The levered beta for the private firm will be calculated using its target debt ratio:

$$1.33 [1 + (1 - 0.40) (0.9)] = 2.048. \text{ Cost of Capital; Section 4.}$$

Question 12

A portfolio manager generated a rate of return of 10% on a portfolio with beta of 1.2. If the risk-free rate of return is 2% and the market return is 8%, Jensen's alpha for the portfolio is closest to:

- A. 0.5%.
- B. 0.8%.**
- C. 1.1%.

B is correct. Jensen's alpha = $R_p - [R_f + \beta_p(R_m - R_f)]$ Jensen's alpha = $0.10 - [0.02 + 1.2 \times (0.08 - 0.02)] = 0.008 = 0.8\%$. Portfolio Risk and Return Part II; Section 4.3.

Question 13

A drag on liquidity is least likely when there are:

- A. high levels of bad debt expenses.
- B. lower number of days of receivables.**
- C. tighter credits because of expensive short-term debt.

B is correct. Lower number of days of receivables means that the company is collecting its receipts rapidly which will not cause a drag on liquidity. Sources of Capital

Question 14

Ahmed Nasri, a portfolio manager, works for wealth management department of an international bank, where he advises high-net-worth clients. He is meeting with a new client, a 45-year-old lawyer with USD 5 million across various accounts and income of USD 550,000 per annum. The client has four accounts at four different banks, each with specific sources and uses of funds.

Bank Account	Source of Deposit	Use of Funds
1	Salary	Living Expenses
2	Bonus	Vacation and Charity
3	Portfolio Interest	Savings for Retirement
4	Portfolio Dividends	Parent's Living Expenses

Based on the above information regarding fund management by the client, he *most likely* exhibits the behavioral bias of:

- A. mental accounting.**
- B. availability.
- C. conservatism.

A is correct. The client has separated money into four different accounts based on the sources and uses of his funds. This method is problematic and contradicts rational economic thought because money is inherently fungible (interchangeable). *The Behavioral Biases of Individuals*.

Question 15

Which of the following is not a primary responsibility of a board's compensation committee?

- A. Set remuneration policies for directors and key executives.
- B. Ensure good corporate governance practices.**
- C. Develop and oversee the implementation of employee benefit plans.

B is correct. Ensuring good corporate governance practices is the responsibility of the Governance Committee. A and C are incorrect because they both fall under the domain of compensation or remuneration committee. Corporate Governance and ESG – An Introduction; Section 5.3.

Question 16

Which of the following is least likely accurate about insurable risks?

- A. Insurable risks are typically diversifiable by the insurer.
- B. Insurable risks are more costly.
- C. Insurable risks have smaller loss limits.**

C is correct. Insurance works by pooling risks. It does not necessarily have lower loss limits. Risk Management – An Introduction; Section 5.3.

Question 17

A project has the following annual cash flows:

Year	0	1	2	3
Cash Flow (\$)	-1000	300	603	305

Which discount rate *most likely* gives a negative net present value?

- A. 7%.
- B. 9%.
- C. 11%.**

C is correct.

Using a financial calculator, enter the following cash flows:

CF0 = -1000, CO1 = 300, CO2 = 603 and CO3 = 305. Press NPV, enter the discount rates, and press CPT NPV.

Discount Rate	NPV (\$)
7%	56.03
9%	18.28
11%	-17.31

Another approach to solving this question would be finding the IRR for the project. If the discount rate is greater than the IRR, the NPV would be negative.

CF0 = -1000, CO1 = 300, CO2 = 603 and CO3 = 305. Press IRR and then press CPT. the IRR is 10.01%. Since 11% is the only option greater than 10.01%, C is the answer. *Uses of Capital*

Question 18

Which of the following is the recommended approach for dealing with flotation costs?

- A. Incorporate flotation costs in the cost of capital.
- B. Adjust cash flows with the amount of the flotation costs.**
- C. Ignore flotation costs.

B is correct. Ideally flotation costs should be incorporated as an adjustment to the cash flows in the valuation computation. Cost of Capital–Foundational Topics

Question 19

Which of the following is least likely a secondary source of liquidity?

- A. Centralized cash management system.**
- B. Renegotiating debt agreements.
- C. Liquidating short term assets.

A is correct. Centralized cash management is a primary source of liquidity. Sources of Capital

Question 20

Momentum Oscillators are *most likely* used to:

- A. set the target price.
- B. indicate an overbought or oversold position.**
- C. analyze the movement of price of security with respect to economic changes.

B is correct. The most known use of momentum oscillators is to indicate the overbought or oversold position of a security. Thus, it helps in providing signal for buying or selling security but it does not help to set the target price. Economic conditions are not used in technical analysis. *Technical Analysis; Section 3.4.*

Question 21

A firm's estimated costs of debt, preferred stock, and common stock are 10%, 15%, and 20%, respectively. Assuming equal funding from each source and a marginal tax rate of 30%, the weighted average cost of capital (WAAC) is closest to:

- A. 12.67%.
- B. 14%.**
- C. 15%.

B is correct. $WACC = w_{drd}(1 - t) + w_{prp} + w_{ere} = [0.10 \times (1 - 0.30) + 0.15 + 0.20]/3 = 14\%$.

A is incorrect, because tax effect is incorrectly calculated $[0.10 \times 0.30 + 0.15 + 0.20]/3 = 12.67\%$.

C is incorrect, because tax effect is ignored $[0.10 + 0.15 + 0.20]/3 = 15\%$. Cost of Capital; Section 2.

Question 22

An ice cream company previously only sold vanilla flavored ice creams. The launch of raspberry ice cream decreased the demand for vanilla ice creams. This effect is most likely known as a:

- A. positive externality.
- B. cannibalization.**
- C. sunk cost.

B is correct. Cannibalization is a primary negative externality which occurs when an investment takes revenue away from another part of the company. Uses of Capital

Question 23

Last year, a portfolio manager earned a return of 15%. The portfolio's beta was 1.3. For the same period, the market return was 9% and the average risk-free rate was 5%. Jensen's alpha for this portfolio is closest to:

- A. 10.2 percent.
- B. 4.8 percent.**
- C. -4.8 percent.

B is correct. $Jensen's\ alpha = 0.15 - [0.05 + 1.3(0.09 - 0.05)] = 0.048$ or 4.8%. Portfolio Risk and Return Part II; Section 4.3.

Question 24

A fifteen-year USD1,000 fixed rate non-callable bond with 7% annual coupons currently sells for USD1,085.45. Assuming a 35% marginal tax rate and an additional risk premium for equity relative to debt of 3%, the cost of equity using the bond-yield-plus-risk-premium approach is closest to:

- A. 8.06%.
- B. 9.11%.**
- C. 10.05%.

B is correct. First, you need to determine the yield-to-maturity, which is the discount rate that sets the bond price to USD1,085.45. This can be done with a financial calculator: FV = -1,000, PV = 1,085.45, N = 15, PMT = -70, solve for I, which will equal 6.11%. The bond-yield-plus-risk-premium approach is calculated by adding a risk premium to the cost of debt (i.e. the yield-to-maturity for the debt) making the cost of equity 9.11% (= 6.11% + 3.00%). Cost of Capital; Section 3.

Question 25

Consider the following income statement of a company:

	\$ (millions)
Revenue	21.8
Variable operating costs	8.9
Fixed operating costs	10.5
Operating income	2.4
Interest	0.9
Taxable income	1.5
Tax	0.8
Net income	0.7

The company's degree of financial leverage is closest to:

- A. 2.5.
- B. 1.6.**
- C. 1.9.

B is correct. $DFL = [2.4 / (2.4 - 0.9)] = 1.6$. Measures of Leverage; Section 3.4.

Question 26

Which one of the following statements is *most accurate* about revolvers?

- A. They are the least reliable form of short-term bank borrowing.
- B. They are in effect for multiple years.**
- C. They often used for much smaller amounts.

B is correct. Revolving credit agreements (Revolvers) are the most reliable form of short-term bank borrowing. They involve formal agreements like those used for regular lines of credit. Revolvers differ from regular lines in two ways (1) they are in effect for multiple years and (2) they are often used for much larger amounts. *Sources of Capital*.

Question 27

An analyst has made the following return projections. Each of the three possible outcomes have an equal likelihood of occurrence.

Stock	Outcome 1 (%)	Outcome 2 (%)	Outcome 3 (%)	Expected Return (%)
ABC	10	5	0	5
LMN	10	0	5	5
XYZ	0	5	10	5

A perfectly negatively correlated portfolio would most likely consist of stocks:

- A. ABC and LMN.
- B. ABC and XYZ.**
- C. LMN and XYZ.

B is correct. Stocks ABC and XYZ have returns that are the same for outcome 2, but the exact opposite returns for outcome 1 and outcome 3. Since they move in opposite directions at the same magnitude, they are perfectly negatively correlated. Portfolio Risk and Return Part I; Section 2.3.

Question 28

Leah's savings accumulate to USD30,000. She borrows another USD30,000 from her broker and invests the total of USD60,000 in a stock, the current price of which is USD25 per share. If the maintenance margin is 35 percent, the first margin call is most likely to occur at:

- A. USD19.23.**
- B. USD12.50.
- C. USD16.25.

A is correct. A margin call will first occur at a price of USD19.23. Because Leah contributed half and borrowed the remaining half, her initial equity is 50 percent of the initial stock price, or $0.50 * USD25 = USD12.5$. If P is the subsequent price, Leah's equity would change by an amount equal to the change in price. So, her equity at price P would be $12.5 + (P - 25)$. A margin call will occur when the percentage margin drops to 35 percent. The price at which a margin call is calculated as follows: $(\text{equity}/\text{share}) / (\text{price}/\text{share}) = 35\%$. Hence, $[(12.5 + P - 25) / P] = 35\%$. $P = USD19.23$. Market Organization and Structure; Section 5.2.

Question 29

Power Bank would like to use some credit enhancement provisions to reduce the credit risk for their next bond issue. The bank has two conditions regarding the use of credit enhancements:

1) A downgrade of the credit provider should not have a negative impact on the bond issue backed by the provider.

2) There should be at least one internal credit enhancement provision.

Which of the following combinations of credit enhancement provisions most likely meets the bank's requirements?

- A. Credit tranching and cash collateral account.**
- B. Surety bonds and cash collateral account.
- C. Letter of credit and credit tranching.

A is correct. Credit tranching is an internal credit enhancement technique which meets condition number 2. A cash collateral account is an actual deposit of cash rather than a pledge of cash. A downgrade of the cash collateral account provider will not necessarily result in a downgrade of the bond issue backed by the provider – meets condition number 1. B is incorrect because condition number 2 is not met. Surety bonds and cash collateral accounts are both external credit enhancement techniques. C is incorrect because condition number 1 is not met. A downgrade of the provider of the letter of credit will negatively impact the bond issue. Fixed Income Securities – Defining Elements; Section 3.1.

Question 30

Which of the following index methods is most likely to result in a value tilt?

- A. Fundamental weighting.**
- B. Equal weighting.
- C. Market-capitalization weighting.

A is correct. Fundamental weighting leads to indices that have a value tilt. Security Market Indices; Section 3.2.

Question 31

Which of the following statements about sovereign bonds is least likely accurate?

Statement I: Sovereign bonds issued in local currency have a higher rating than those issued in a foreign currency.

Statement II: A national government has the ability to print currency and hence, it is not restricted in being able to service its sovereign bonds in any currency.

Statement III: Sovereign bonds are not backed by any collateral but by the taxing authority of the national government.

- A. Statement I.
- B. Statement II.**
- C. Statement III.

B is correct. A national government has the ability to print its own currency, whereas it is restricted in being able to pay in a foreign currency only what it earns in exports or can exchange in financial markets. Thus, it is restricted in its ability to service debt in a foreign currency. Statements I and III are correct. Fixed Income Markets – Issuance Trading and Funding; Section 4.

Question 32

A hedge fund strategy that entails advocating restructuring, changes in strategy or hiving off non-profitable units is:

- A. short bias.
- B. activist.**
- C. multi-strategy.

B is correct. Activist strategy involves purchasing a managing equity stake in a public company that is believed to be mismanaged, and then influencing its policies.

Short bias strategies use quantitative and/or fundamental analysis to identify overvalued securities.

Multi-strategy focusses on generating consistently absolute positive returns irrespective of how the equity, debt, or currency markets are performing. *Introduction to Alternative Investments*.

Question 33

Which of the following statements about forward contracts is most likely correct?

- A. A forward contract requires the backing of a clearinghouse.
- B. Forward contracts tend to trade in organized markets.
- C. A forward contract can be used to eliminate uncertainty of the future price of an asset.**

C is correct. In a forward contract, one party agrees to buy and the counterparty agrees to sell an asset at a specific price on a specific date in the future hence eliminating the price uncertainty about the future price of an asset it plans to buy or sell at a later date. Forward contracts are private contracts that do not trade on an exchange and do not trade in organized markets. Derivative Markets and Instruments; Section 4.1.

Question 34

Chess Inc.'s stock currently trades at a price of USD50 per share. It is expected that after an announcement, the price would rise to USD52, but it rises up to USD55. This anomaly is most likely known as:

- A. earnings surprise.
- B. January effect.
- C. overreaction effect.**

C is correct. This anomaly is most likely known as the overreaction effect. Market Efficiency; Section 4.1.

Question 35

An analyst wants to value an annual coupon pay bond maturing 3 years from now and pays a 5% coupon. The par value of the bond is USD100. Current spot rate is 2%, 1-year forward rate 1 year from now is 3% and 1-year forward rate 2 years from now is 4%. The value of this annual coupon pay bond is closest to:

- A. USD102.96.
- B. USD98.89.
- C. USD105.76.**

C is correct. Bond value = $5/(1+S_1) + 5/((1+S_1)(1+1y1y)) + 105/((1+S_1)(1+1y1y)(1+2y1y))$
Bond value = $5/1.02 + 5/((1.02)(1.03)) + 105/((1.02)(1.03)(1.04)) = \text{USD}105.76$. Introduction to Fixed Income Valuation; Section 2.4.

Question 36

Which of the following transactions could potentially have the highest loss?

- A. Selling a call option.**
- B. Selling a put option.
- C. Buying a put option.

A is correct. The profit potential to the buyer of the call option is unlimited and hence, the potential loss to the seller of the call option is unlimited. Option trading is a zero-sum game. The

maximum loss for the seller of a put option is strike price minus option premium and for a buyer of a put option is the option premium. Basics of Derivative Pricing and Valuation; Section 4.

Question 37

Compared to investing directly in hedge funds, an investor would most likely prefer investing in fund of hedge funds because:

- A. of a simpler fee structure.
- B. of unfavorable terms of redemption.
- C. of diversified exposure to hedge fund strategies.**

C is correct. A fund of hedge funds offers advantages such as access by smaller investors, diversified hedge fund portfolio, better redemption terms, and/or due diligence expertise that multi-strategy funds do not have. A primary difference between a multi-strategy hedge fund and a fund of hedge funds is the extra layer of fees associated with a fund of funds. Each hedge fund in which a fund of hedge funds invests is structured to receive a management fee plus an incentive fee, and the fund of hedge funds is also structured to receive a management fee and may receive an incentive fee. Introduction to Alternative Investments; Section 3.3.

Question 38

Which of the following is most likely a reason for companies to issue equity in primary markets?

- A. To increase return on equity.
- B. To increase return on capital.
- C. To raise capital.**

C is correct. A company issues equity in primary markets to raise capital and increase liquidity. Overview of Equity Securities; Section 7.

Question 39

Which of the following statements about a special purpose vehicle (SPV) is most accurate?

- A. The major advantage of a SPV is bankruptcy remoteness.**
- B. The major advantage of a SPV is beneficial tax treatments.
- C. The major advantage of a SPV is lower issuing costs.

A is correct. An SPV is a bankruptcy-remote vehicle. If the sponsor defaults, no claims can be made to recover the assets that were transferred to the SPV. Introduction to Asset Backed Securities; Section 3.4.

Question 40

An analyst has gathered the following data:

- Market value of debt: USD20 million
- Market capitalization: USD55 million
- Cash and short-term investments: USD7.5 million
- EBITDA: USD18 million
- Firm's marginal tax rate: 32%

The company's EV/EBITDA multiple is closest to:

- A. 3.85.
- B. 3.75.**
- C. 3.38.

B is correct. Enterprise value (EV) = Market capitalization + MV of debt + MV of preferred stock – Cash and short-term investments.

$EV = 55 + 20 + 0 - 7.5 = 67.5$; $EV/EBITDA = 67.5/18 = 3.75$. Equity Valuation; Section 5.4.

Question 41

Which of the following statements about convexity of a bond is most likely correct?

- A. An option-free bond has greater convexity than an otherwise identical putable bond.
- B. The convexity of a callable bond can be negative at low yields.**
- C. A bond with a high yield to maturity will have greater convexity than a bond with low yield to maturity.

B is correct. The convexity of a callable bond can be negative at low yields because at low yields the call option becomes more valuable and the call price puts an effective limit on increases in bond value. Statement A is incorrect because a putable bond has greater convexity than an otherwise identical option-free bond. Statement C is incorrect because a low yield to maturity bond will have greater convexity than a high yield to maturity bond. Understanding Fixed-Income Risk and Return; Section 3.4.

Question 42

Analyst 1: Derivative markets exhibit lower volatility as compared to spot markets.

Analyst 2: Derivative markets enable companies to practice risk management more easily.

Which of the following is most likely correct?

- A. Only Analyst 1 is correct.
- B. Only Analyst 2 is correct.**
- C. Both Analysts are correct.

B is correct. One of the main purposes of derivative markets is to make risk management easier. Derivative markets are not necessarily more or less volatile as compared to spot markets. Hence A and C are incorrect. Derivative Markets and Instruments; Section 5.

Question 43

An investor is least likely to consider adding alternative investments to a traditional investment portfolio because of their:

- A. higher Sharpe ratio.
- B. low correlation with traditional investments.
- C. historically higher standard deviation of returns.**

C is correct. Historically higher standard deviation of returns of most categories of alternative investments can be one of the reasons why people may not invest in them. Introduction to Alternative Investments; Section 2.1.

Question 44

Which of the following is most likely a financial asset?

- A. Currencies.**
- B. Commodities.
- C. Real estate.

A is correct. Currencies are financial assets. Market Organization and Structure; Section 3.1.

Question 45

The default and payment acceleration of cash flows is delayed until a new maturity date in case of:

- A. Hard bullet covered bonds
- B. Soft bullet covered bonds**
- C. Conditional pass-through covered bonds

B is correct. Soft bullet covered bonds delay the default and payment acceleration of cash flows until a new maturity date, which is generally up to a year after the original maturity date. In case of hard bullet covered bonds, if payments do not occur according to the original schedule, a default is triggered and payments are expedited. Conditional pass-through covered bonds convert to pass-through securities after the original maturity date if all bond payments have not yet been made. *Introduction to Asset Backed Securities*.

Question 46

A futures contract:

- A. can be settled only by delivery.
- B. has a value of zero at initiation of the contract.**
- C. is a customized contract that satisfies the specific needs of the parties involved.

B is correct. Futures contracts can be either deliverable or cash-settled contracts. Also, futures contracts are standardized contracts that specify the quality and quantity of assets required under the contract and the delivery procedure for deliverable contracts. *Derivative Markets and Instruments*; Section 4.1.

Question 47

Which of the following is most likely a disadvantage of IPO as an exit strategy for private equity firms?

- A. Opposition by management.
- B. Lower levels of disclosure requirements.
- C. High transaction costs.**

C is correct. Disadvantages for an IPO exit include high transaction costs paid to investment banks and lawyers and high disclosure requirements. Management approval is an advantage of IPO exit strategy as they are retained after the company goes public. *Introduction to Alternative Investments*; Section 4.2.

Question 48

An analyst gathers the following information for an equal-weighted index comprised of three securities:

Security	Number of shares	Beginning of period price (\$)	End of period price (\$)	Total Dividend (\$)
Alpha	100	13	13	3
Beta	350	15	20	6

Gamma	500	25	16	2
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The price return of the index is closest to:

- A. – 14.4 percent.
- B. – 7.6 percent.
- C. – 0.9 percent.**

C is correct. The price return of the index equals the weighted average of price returns of the individual securities.

Return of Alpha: $[(13 - 13)/13] = 0$ percent.

Return of Beta: $[(20 - 15)/15] = 33.3$ percent.

Return of Gamma: $[(16 - 25)/25] = -36$ percent.

The price return index assigns equal weights to each asset; therefore, the price return is $1/3 \times [(0\%) + (33.3\%) - (36\%)] = -0.9\%$. Security Market Indices; Section 2.

Question 49

The repo rate on a repurchase agreement is most likely lower:

- A. for long term agreements as compared to short term agreements.
- B. when delivery to the lender is not required.
- C. if the underlying security is in great demand.**

C is correct. If the security is in great demand the borrower will pay a lower repo rate on cash borrowed. Statement A is incorrect because repo rates generally increase with maturity because long-term rates are typically higher than short-term rates in normal circumstances. Statement B is incorrect because repo rates are usually lower when delivery to the lender is required. Fixed Income Markets – Issuance Trading and Funding; Section 8.3.

Question 50

Which of the following statements is least likely a limitation to exploit an arbitrage opportunity?

- A. Most investors are risk averse.**
- B. Transaction costs.
- C. Ability to borrow funds.

A is correct. If the arbitrage profit opportunity is very small, it is possible that the transactions costs might exceed the arbitrage profit making it an unprofitable transaction. Ability to borrow funds is also one of the limitations to exploit an arbitrage opportunity as one cannot borrow unlimited funds to create new or maintain existing positions in the market. Investor's risk aversion behavior is not a limitation to exploit arbitrage opportunities. Basics of Derivative Pricing and Valuation; Section 2.3.

Question 51

An analyst that is considering risk management measures for alternative investments is *least likely* to consider:

- A. Calmar ratio.
- B. Sharpe ratio.**
- C. Batting average.

B is correct. Traditional risk and return measures such as the Sharpe ratio (measure of return per unit of risk) are not always appropriate for alternative investments because such investments exhibit asymmetric risk and return profile, which means they might have high kurtosis (leptokurtic) and negative skewness. *Introduction to Alternative Investments*.

Question 52

In the event of board elections of a company, the type of voting that will be most beneficial to shareholders with a small number of shares is best described as:

- A. **cumulative voting.**
- B. statutory voting.
- C. voting by proxy.

A is correct. Cumulative voting allows shareholders to direct their total voting rights to specific candidates, as opposed to having to allocate their voting rights evenly among all candidates. Thus, applying all of the votes to one candidate provides the opportunity for a higher level of representation on the board than would be allowed under statutory voting. *Overview of Equity Securities; Section 3.1*.

Question 53

Given the following information, the estimated spread on a newly issued 5-year, BBB rated corporate is closest to:

Issue	YTM
4-year, US Treasury bond	2.48%
4-year, BBB rated corporate bond	3.65%
6-year, US Treasury bond	3.00%
6-year, BBB rated corporate bond	4.58%
5-year, US Treasury bond	2.73%

- A. 2.73%.
- B. **1.38%.**
- C. 4.13%.

B is correct. Spread on the 4-year corporate bond is $3.65\% - 2.48\% = 1.17\%$ and spread on the 6-year corporate bond is $4.58\% - 3.00\% = 1.58\%$. 5-year corporate bond is the midpoint of four and six years hence, average spread is $(1.17\% + 1.58\%)/2 = 1.38\%$. *Introduction to Fixed Income Valuation; Section 5*.

Question 54

The economic profit is most likely to be larger for an industry with:

- A. **high switching costs to customers.**
- B. low pricing power.
- C. standardized products.

A is correct. The economic profit (the spread between the return on invested capital and the cost of capital) tends to be larger in industries with differentiated products, greater pricing power, and high switching costs to customers. *Introduction to Industry and Company Analysis; Section 5*.

Question 55

Consider a CMO structure with one planned amortization class and one support tranche. The initial PAC collar was 100-250 PSA. If the actual prepayment speed is 300 PSA, the average life of the PAC tranche will:

- A. **contract.**
- B. extend.
- C. remain the same.

A is correct. If the actual PSA rate is higher than the upper collar of the PAC, the average life of the PAC tranche will contract. Introduction to Asset Backed Securities; Section 5.2.

Question 56

Based on the put call parity relation, which of the following is accurate?

- A. The payoff on a long stock can be created with a long call, a long put and a short position in a risk-free bond.
- B. The payoff on a long put can be created with a long call, a short position in a risk-free bond and a long stock.
- C. **The payoff on a long call can be created with a long put, a long stock and a short position in a risk-free bond.**

C is correct. The payoff on a long stock can be created with a long call, a short put and a long position in a risk-free bond. The payoff on a long put can be created with a long call, a long position in a risk-free bond and a short stock. Basics of Derivative Pricing and Valuation; Section 4.1.

Question 57

An investor seeks the potential for competitive long-term total returns driven by both income generation and capital appreciation. The investment most likely to achieve the investor's goals is:

- A. private equity.
- B. commodities.
- C. **real estate.**

C is correct. One of the key reasons for investing in real estate is the potential for competitive long-term total returns driven by both income generation and capital appreciation. Introduction to Alternative Investments; Section 5.

Question 58

The following information on a company is provided:

- Earnings per share (2012): USD2.00
- Dividend payout ratio (2012): 50%
- Dividend growth rate expected during Years 2013 and 2014: 10%
- Dividend growth rate expected after Year 2014: 5%
- Investors' required rate of return: 12%

Using the two-stage dividend discount model, the value per share of this common stock at the end of 2012 is closest to:

- A. USD15.21.
- B. USD16.05.
- C. **USD16.41.**

C is correct. $DPS(2012) = 2.00 \times 50\% = 1.00$

$V_{2012} = 1(1.1)/1.12 + 1(1.1)^2/1.122 + 1(1.1)^2(1.05)/(0.12 - 0.05) \times 1/1.122$

$V_{2012} = 0.98 + 0.96 + 14.47 = USD16.41$. Equity Valuation; Section 4.5.

Question 59

Which of the following statements about market liquidity risk is most likely accurate?

- A. Less outstanding debt for an issuer results in lower market liquidity risk.
- B. Less outstanding debt for an issuer results in higher market liquidity risk.**
- C. Lower quality of the issuer results in lower market liquidity risk.

B is correct. The two main issuer-specific factors that affect market liquidity risk are (1) the amount of publicly traded debt an issuer has outstanding and (2) the credit quality of the issuer. In general, the less debt an issuer has outstanding, the less frequently its debt trades, and thus the higher the market liquidity risk. And the lower the quality of the issuer, the higher the market liquidity risk. Fundamentals of Credit Analysis; Section 2.

Question 60

Which of the following statements best describes an advantage of a forward contract over a futures contract?

- A. A forward contract is default free.
- B. A forward contract allows parties to enter into a customized transaction.**
- C. A forward contract can easily be offset prior to expiration.

B is correct. Unlike futures contracts, which have standardized features, forward contracts can be customized to suit the needs of the parties involved. Derivative Markets and Instruments; Section 4.1.

Question 61

A real estate investment fund has a USD100 million initial drawdown structure in its first year and fully draws this capital to purchase a property. The fund has a soft hurdle preferred return to investors of 8% per annum and an 80%/20% carried interest incentive split thereafter. At the end of year two, the property is sold for a total of USD180 million. Total distribution to GP would be:

- A. 25.6 million**
- B. 12.8 million
- C. 38.4 million

A is correct.

The waterfall of cash flows would be constructed as follows.

- First, the LPs would be due their USD100 million initial investment.
- Then, they would be due USD16 million (8% preferred return on initial capital for two years).
- The soft hurdle has been met, and the GP is ultimately due 20% of USD80 million, or USD16 million, which would be paid to the GP next as a catch-up to the achieved hurdle return.
- The residual amount would be USD180 million – USD100 million – USD16 million – USD16 million = USD48 million. This amount would then be split 80% to the LPs and 20% to the GP, or 38.4millionand 9.6 million, respectively.

Total payout with a soft annual hurdle of 8% of the \$180 million would end up with the following waterfall:

	LP	GP
Return of Capital	100.00	
8% Prfd per Annum	16.00	
GP Catch-Up 20%		16.00
80%/20% Split	38.40	9.60
	154.40	25.60

Introduction to Alternative Investments

Question 62

Sofia Santos buys a stock priced at \$50 on margin with a leverage ratio of 2.5 and a maintenance margin of 20%. She will most likely receive a margin call when the stock price is at or falls just below:

- A. \$25.0.
- B. \$32.0.
- C. \$37.5.**

C is correct. $\text{Equity} = 1/\text{Leverage ratio} = 1/2.5 = 0.4$

Initial equity = $50 * 0.4 = \text{USD}20$. Price (P) for margin call: $(20 + P - 50) / P = 20\%$; $P = \text{USD}37.5$.

Alternatively: $P = 50 (1 - 0.4) / (1 - 0.2) = 37.5$. Market Organization and Structure; Section 5.2.

Question 63

Compared to its fundamentally weighted counterpart, a market-value-weighted index will most likely have a:

- A. contrarian effect.
- B. momentum effect.**
- C. value tilt.

B is correct. "Momentum effect" is a characteristic of a market-value-weighted index, not a fundamentally weighted index. Security Market Indices; Section 3.2.

Question 64

In an underwritten offering, the investment bank:

- A. serves as a broker.
- B. takes the risk associated with selling the bond.**
- C. has less risk than in a best effort offering.

B is correct. In an underwritten offering, the investment bank guarantees the sale of the bond issue at an offering price that is negotiated with the issuer. Thus, the investment bank, called the underwriter, takes the risk associated with selling the bonds. In contrast, in a best effort offering, the investment bank only serves as a broker. It only tries to sell the bond issue at the negotiated offering price if it is able to for a commission. Thus, the investment bank has less risk and correspondingly less incentive to sell the bonds in a best effort offering than in an underwritten offering. Fixed Income Markets – Issuance Trading and Funding; Section 3.1.

Question 65

Acme Investment Ltd. is a fixed-income investment firm that actively manages a government bond fund and a corporate bond fund. Government bond fund is mainly invested in medium-term Treasury securities and highly rated developed-market sovereign issues. The corporate bond fund is invested in high-yield issues from North American, Asian and African markets.

Which of the following is an appropriate risk measure for Acme's corporate fund compared to its government bond fund?

- A. **Empirical duration.**
- B. Analytical duration.
- C. None of the above.

A is correct. The corporate bond fund includes a wide variety of high-yield bonds with different credit and liquidity spreads. For high yield bonds with significant credit risk, the analytical and empirical duration will be different. In a market stress scenario, many investors switch to high quality government bonds due to which their yields (i.e., benchmark yields) fall. But at the same time the credit spreads on high-yield bonds will widen (i.e., credit spreads and benchmark yields are negatively correlated). The wider credit spreads will fully or partially offset the decline in government benchmark yields. Thus, the empirical duration for high yield bonds will be lower than their analytical duration. Hence, empirical duration is a better risk measure for the corporate bond fund.

For the government bond with little or no credit risk, the analytical and empirical duration would be similar because bond prices are largely driven by changes in the benchmark yield.

Understanding Fixed Income Risk and Return

Question 66

Security markets are generally not strong-form efficient because:

- A. **insider trading is restricted by regulations.**
- B. prices automatically reflect any new information in the market
- C. change in prices cannot be linked to current information in the market.

A is correct. Security markets are not strong-form market efficient because insider trading is restricted. Market Efficiency; Section 3.3.

Question 67

Mike Weise, CFA, is following a Panache & Co. 6.25%, 3-year, semiannual-pay bond trading at 102.63% of par value. Assuming par value is USD100 and the bond is callable at USD102 in two years, what is the bond's yield-to-call?

- A. 5.481%.
- B. 2.898%.
- C. **5.796%.**

C is correct. $N = 4$; $FV = 102$; $PMT = 6.25/2$; $PV = -102.63$; $CPT\ I/Y = 2.898 \times 2 = 5.796\%$.
Introduction to Fixed Income Valuation; Section 3.3.

Question 68

Which of the following *best describes* the value of a forward contract to a long party?

- A. The value of a forward contract at initiation benefits the long party in the contract.
- B. The value of a forward contract at expiration is the agreed forward price minus the spot price.
- C. The value of a forward contract is the spot price of the underlying asset minus the present value of the forward price.**

C is correct. Statement A is incorrect because neither the long nor the short pays anything to the other at the initiation date of a forward contract, the value of a forward contract when initiated is zero. Statement B is incorrect because the value of a forward contract at expiration is the spot price of the underlying minus the forward price agreed to in the contract. *Basics of Derivative Pricing and Valuation; Section 3.1.*

Question 69

When futures prices are lower than the spot price, the commodity forward curve is downward sloping and prices are referred to as being in:

- A. contango.
- B. equilibrium.
- C. backwardation.**

C is correct. A downward sloping curve occurs when the futures prices are lower than the spot price. This condition is called backwardation. Opposite is true for contango. *Introduction to Alternative Investments*

Question 70

A UK-based investor who wishes to invest in depository instruments of non-domestic equity securities which are not subject to the foreign ownership and capital flow restrictions will most likely opt for a(an):

- A. index-based hedge fund.
- B. global depository receipts.**
- C. American depository receipts.

B is correct. Global depository receipts (GDRs) meet the investor preferences. They are not subject to the foreign ownership and capital flow restrictions that may be imposed by the issuing company's home country because they are sold outside of that country. *Overview of Equity Securities; Section 5.2.*

Question 71

Based on the following ABS structure, what will be the losses to Tranche B in case of a USD35,000,000 loan default?

- Senior Tranche: USD200,000,000
 - Subordinated Tranche A: USD25,000,000
 - Subordinated Tranche B: USD25,000,000
 - Collateral value: USD285,000,000
- A. \$0 because senior tranche will absorb the entire loss amount.
 - B. \$25,000,000 because Tranche B will be the first to absorb losses.
 - C. \$0 because the entire loss amount will be absorbed by overcollateralization.**

C is correct. Overcollateralization = USD285,000,000 – USD250,000,000 = USD35,000,000. This means that losses up to USD35,000,000 will be absorbed first by overcollateralization and none of the tranches will experience a loss. Introduction to Asset Backed Securities; Section 8.

Question 72

Which of the following industries is *least likely* to be characterized as concentrated with weak pricing power?

- A. Commercial aircraft.
- B. Automobiles.
- C. Credit Card Networks**

C is correct Credit card networks will be classified as a concentrated industry with strong pricing power. A and B are examples of concentrated industries with weak pricing power. *Industry and Company Analysis*.

Question 73

Company A's debt was rated BBB by credit rating agencies last year. However, recently company A's debt rating has moved down to junk bonds. The least likely reason for this downgrade could be:

- A. that the company issued more debt to finance its operations.
- B. the company's reduced sensitivity to business cycles.**
- C. that the company acquired some large off-balance-sheet liabilities to run its business.

B is correct. Some of the reasons for credit rating agencies to downgrade a company's debt to junk are: high leverage, high sensitivity to business cycles and large off-balance-sheet liabilities. Hence, a company's reduced sensitivity to business cycles is not a reason for credit rating agencies to downgrade corporate debt. Fundamentals of Credit Analysis; Section 4.

Question 74

Consider the following information regarding a company's stock:

- Risk-free rate: 5 percent.
- Market risk premium: 7 percent.
- Stock's beta: 1.5
- Company's ROE: 15 percent.
- Expected rate of return: 11 percent.
- Company's weighted average cost of capital: 13 percent.

Based on this, an investor should most likely:

- A. invest in the stock.
- B. not invest in the stock as WACC is greater than required rate of return.
- C. not invest in the stock.**

C is correct. Stock's required rate of return according to CAPM = risk-free rate + Beta*market risk premium

$$0.05 + 1.5 \times 0.07 = 0.155 \times 100 = 15.5\%$$

Since the stock's required rate of return (15.5%) is higher than the expected rate of return (11%), investor should not invest in this stock. Equity Valuation; Section 4.2.

Question 75

If a bond is callable any time after the call protection period, it is most likely a(n):

- A. American style callable bond.**
- B. European style callable bond.
- C. Bermuda style callable bond.

A is correct. Under American style of exercise, a callable bond can be called any time after the call protection period. A European callable bond can be called only once on the call date. A Bermuda style callable bond can be called on predetermined dates following the call protection period. Fixed Income Securities – Defining Elements; Section 5.1

Question 76

Other things being equal, at expiration an American option and a European option:

- A. have different values.
- B. have the same value.**
- C. have to be exercised.

B is correct. At expiration, an American option and a European option on a particular asset with the same strike price are identical. They may either be exercised or allowed to expire. They are different and have different values before expiration. Basics of Derivative Pricing and Valuation; Section 4.3.

Question 77

Giyani Fund of Funds invests EUR150 million in each of Beta Hedge Fund and Delta Hedge Fund. Giyani FOF has a “2 and 15” fee structure. Management fees and incentive fees are calculated independently at the end of each year. After one year, net of their respective management and incentive fees, the investment in Beta is valued at EUR120 million and the investment in Delta is valued at EUR210 million. The annual return to an investor in Giyani, net of fees assessed at the fund of funds level, is closest to:

- A. 6.30%.**
- B. 8.63%.
- C. 10.00%.

A is correct. The net investor return is 6.3%, calculated as:
 First, note that “2 and 15” refers to a 2% management fee, and a 15% incentive fee.
 End of year capital = EUR120 million + EUR210 million = EUR330 million
 Management fee = EUR330 million × 2% = EUR6.6 million
 Incentive fee = (EUR330 – EUR300) million × 15% = EUR4.5 million
 Total fees = (EUR6.6 + EUR4.5) million = EUR11.1 million
 Investor net return: (EUR330 – EUR300 – EUR11.1) / EUR300 = 6.3%. Introduction to Alternative Investments; Section 3.3.

Question 78

Meredith has sold short 200 shares of Cola Inc. at a price of \$45 per share. At the same time, she also placed a “good-till-cancelled, stop 50, limit 60 buy” order. Assuming no transaction costs, if the stock prices starts to increase rapidly, the maximum possible realized loss is closest to:

- A. \$3,000.**
- B. \$2,000.
- C. \$750.

A is correct. If the stock price crosses USD50, the stop buy order will become valid and will get executed at a maximum limit price of USD60. The maximum loss per share will be USD60 – USD45 = USD15, or $\text{USD15} \times 200 = \text{USD3,000}$ for 200 shares. Market Organization and Structure; Section 6.2

Question 79

Non-negotiable certificates of deposit most likely:

- A. are traded in the open market.
- B. are purchased only by retail investors.
- C. have a penalty for early withdrawal of funds.**

C is correct. With non-negotiable CD, there is a penalty if the depositor withdraws funds before maturity. Fixed Income Markets – Issuance Trading and Funding; Section 8.2.

Question 80

In case of futures contracts, the exchange can provide a credit guarantee because:

- A. futures markets are heavily regulated.
- B. of daily settlement of gains and losses.**
- C. futures contracts are standardized contracts making them easier to mark to market.

B is correct. The futures are marked to market on a daily basis, meaning that the accumulated gains and losses from the previous day's trading session are deducted from the accounts of those holding losing positions and transferred to the accounts of those holding winning positions. This daily settling of gains and losses enables the futures exchange to guarantee that a party that earns a profit from a futures transaction will not have to worry about collecting the money. Thus, futures exchanges provide a credit guarantee. Derivative Markets and Instruments; Section 4.1.

Question 81

Which of the following statements is *most likely* to be correct about funds of funds?

Statement I: Funds of funds are funds that hold a portfolio of hedge funds.

Statement II: Funds of funds presumably have some expertise in conducting due diligence on hedge funds.

Statement III: Funds of funds may be able to negotiate better redemption

- A. Statements I and II.
- B. Statements I and III.
- C. Statements I, II, and III.**

C is correct. All three statements regarding funds of funds are correct. A portfolio of hedge funds is often referred to as a fund of funds. This instrument makes hedge funds accessible to smaller investors. Other benefits include: Better redemption terms. Funds of funds assumed to have some expertise in conducting due diligence on hedge funds. More diversification as they invest in hedge funds across geographies and strategies. *Introduction to Alternative Investments; Section 3.*

Question 82

Observed overreactions in securities market are most likely due to:

- A. mental accounting.
- B. loss aversion.**
- C. narrow framing.

B is correct. Loss aversion arguments suggest that investors dislike losses more than they like comparable gains. This might explain observed overreactions in securities markets. Market Efficiency; Section 5.

Question 83

In case of a residential mortgage, the higher the loan-to-value ratio:

- A. the higher the homeowner's equity.
- B. the more likely the borrower is to default.**
- C. the less likely the borrower is to default.

B is correct. The ratio of the amount of the mortgage to the property's value is called the loan-to-value ratio (LTV). The lower the LTV, the higher the borrower's equity. From the lender's perspective, the higher the borrower's equity, the less likely the borrower is to default. Introduction to Asset Backed Securities; Section 4.

Question 84

An increase in the exercise price will:

- A. increase put and call option prices.
- B. increase put option prices and decrease call option prices.**
- C. increase call option prices and decrease put option prices.

B is correct. A higher exercise price decreases the value of a call option and increases the value of a put option. This is because the probability of being in the money reduces as the strike price increases for call options and the probability of being in the money increases as the strike price increases for put options. Basics of Derivative Pricing and Valuation; Section 4.1.

Question 85

Which of the following hedge fund strategies is *least likely* categorized as an event-driven strategy?

- A. Distressed debt.
- B. Merger arbitrage.
- C. Convertible bond arbitrage.**

C is correct. Event-driven strategies include merger arbitrage, distressed/restructuring, activist shareholder and special situation. Convertible bond arbitrage is classified as a relative value strategy. It is a market-neutral strategy which seeks to exploit mispricing in convertible bond and issuer's stock. *Introduction to Alternative Investments*.

Question 86

The following data on a company is provided:

- Par value of a perpetual, non-convertible, non callable preferred stock offered at a 5% dividend rate: USD100
- Company's sustainable growth rate: 4%

- The required rate of return on similar issues is: 11.5%
- Investor's marginal tax rate: 30%

The value of the company's preferred stock is closest to:

- A. USD43.48.**
- B. USD45.26.
- C. USD48.35.

A is correct. $V_0 = D_0 / r = 100 \times 0.05 / 0.115 = \text{USD}43.48$. Equity Valuation; Section 4.3.

Question 87

The duration and convexity of an option-free bond priced at USD95.50 are 8.50 and 85.00, respectively. If yields increase by 150 bps, the percentage change of the price is closest to:

- A. 8.50%.
- B. 10.84%.
- C. 11.79%.**

C is correct. Duration effect = $-8.50 \times (0.015) = -0.1275$

Convexity effect = $\frac{1}{2} \times 85 (0.015)^2 = 0.00956$

Total percentage change in the price is the sum of duration effect and convexity effect.

Total percentage change in the price = $-0.1275 + 0.00956 = 0.1179 = 11.79\%$. Understanding Fixed-Income Risk and Return; Section 4.1.

Question 88

According to the put-call-forward parity, the difference between the price of put and price of call is most likely equal to:

- A. difference between the exercise price and forward price.
- B. difference between the exercise price and forward price discounted at the risk-free rate.**
- C. sum of the exercise price and forward price discounted at the risk-free rate.

B is correct. The put-call-forward parity can be written as:

$p_0 - c_0 = [X - F_0(T)] / (1 + r)^T$. Basics of Derivative Pricing and Valuation; Section 4.1.

Question 89

An investor considering the enterprise value approach to valuation gathers the following data:

- EBITDA: USD55 million.
- Value of debt: USD70 million
- Value of preferred stock: USD33 million
- Cash & marketable securities: USD14 million
- Number of common shares outstanding: 15 million
- Firm's tax rate: 32%
- Appropriate EV/EBITDA multiple: 6x

The value per share of the company's common stock is closest to:

- A. \$16.07.**
- B. \$20.47.
- C. \$29.80.

A is correct. First, compute the enterprise value (EV) from EBITDA * EV/EBITDA multiple. EV = USD55 million x 6 = USD330 million

Then determine market capitalization (value of equity) using the following expression.

Value of equity = EV – Value of debt – value of preferred stock + cash & marketable securities =
USD330 – USD70 – USD33 + USD14 = USD241 million

Finally, compute the value per share. $\text{USD241} / 15 = \text{USD16.07}$. Equity Valuation; Section 5.4.

Question 90

Caesar Corp. and Warm Salads Corp. are two major players in the retail food and beverage industry. In spite of having similar debt/capital ratios, what is the most likely reason the credit spread of Caesar Corp. is higher than Warm Salads Corp.?

- A. Warm Salads Corp. has a higher debt/EBITDA than Caesar Corp.
- B. Warm Salads Corp. has a lower EB IT margin than Caesar Corp.
- C. Caesar Corp. has a lower FFO/debt ratio than Warm Salads Corp.**

C is correct. A lower value FFO (Funds from operations) to debt ratio implies poor credit quality. Statements A and B indicate that Caesar Corp. should have a lower credit spread as compared to Warm Salads Corp. *Financial Analysis Techniques*